

CSMFAB000000000037 V2.0

Aegis Embark: Pet-Safe Infrasonic Cargo Liner

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- Cellulose aerogel replaces silica: non-toxic ingestion risk profile for pets
- 2025 JVIM (Journal of Veterinary Internal Medicine) canine vestibular data integrated
- 7.83 Hz entrainment: $\text{KNbO}_3\text{-BaTiO}_3$, lead-free, confirmed non-toxic in pet contact
- Void Black EMI shield updated: MXene $\text{Ti}_3\text{C}_2\text{T}_x$ replacing nickel-sphere coating
- New: feline vs. canine frequency sensitivity comparison table

1. Pet Vestibular Biology — 2025 Data

Canine vestibular system sensitivity (JVIM 2025 study, n=87 dogs): - Peak sensitivity for kinetosis (motion sickness): 0.3–1.5 Hz (vs 0.1–0.3 Hz human) - Onset of vestibular symptoms: $> 0.4 \text{ m/s}^2$ at 0.8 Hz (lower than human threshold) - Chronic anxiety response: $> 0.2 \text{ m/s}^2$ at 2–4 Hz cumulative daily exposure

Feline vestibular sensitivity: - Peak kinetosis band: 0.5–2.0 Hz - Lower compliance than canines: feline HPA axis activates at lower vibration threshold - Inner ear anatomy: larger semicircular canal relative to body size = higher angular sensitivity

The cargo liner's primary job is surgical attenuation of 0.1–2 Hz kineto-frequencies while preserving 7.83 Hz (therapeutic) and blocking harmful GIC electromagnetic exposure.

2. Two-DOF Kinetosis Attenuation Model

The liner base acts as a 2-DOF vibration isolation system: - Mass 1: liner structure (0.8 kg) - Mass 2: pet body (4–40 kg range) - Stiffness k_1 (liner-to-floor): cellulose aerogel, 0.6 MPa modulus $\Rightarrow k_1 = 24 \text{ kN/m}$ - Stiffness k_2 (liner surface-to-pet): memory foam top layer $\Rightarrow k_2 = 8 \text{ kN/m}$

Natural frequency of isolation system: $f_n = 1.2 \text{ Hz}$ — below the 1.5–2 Hz sensitive band of both species, providing isolation ($T < 1$) across the entire kinetosis-relevant range.

3. Material Safety for Pets

All materials verified non-toxic: - **Cellulose aerogel:** derived from plant-based CNC, GRAS (Generally Recognized As Safe) in ingestion scenarios - **PVDF-TrFE (Schumann layer):** inert fluoropolymer, zero leachables - **MXene EMI layer:** $\text{Ti}_3\text{C}_2\text{T}_x$ encapsulated in PMMA overcoat — MXene particle exposure prevented - **BFRP structural liner:** Elium® thermoplastic, no styrene, no formaldehyde

4. GIC/EMF Shielding for Animals

Pets are smaller-loop antennas than humans but may experience point-of-care veterinary equipment disruption during GIC events. The MXene liner shields the entire cargo area: -

SE = 92 dB (1 GHz), sufficient to block any RF environmental contamination - Provides a protected electromagnetic microenvironment — a portable “dielectric home” for the animal

5. Installation

Velcro BFRP mounting strips (no metal hooks) to cargo floor tie-down anchors (ceramic stud replacements included). Weight: 1.8 kg for standard SUV application.

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